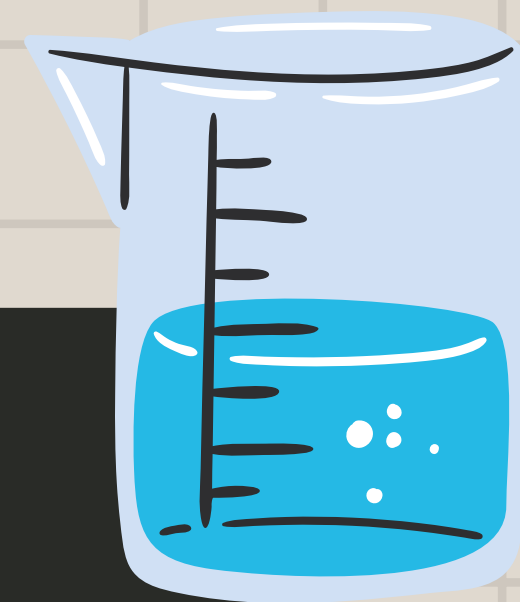
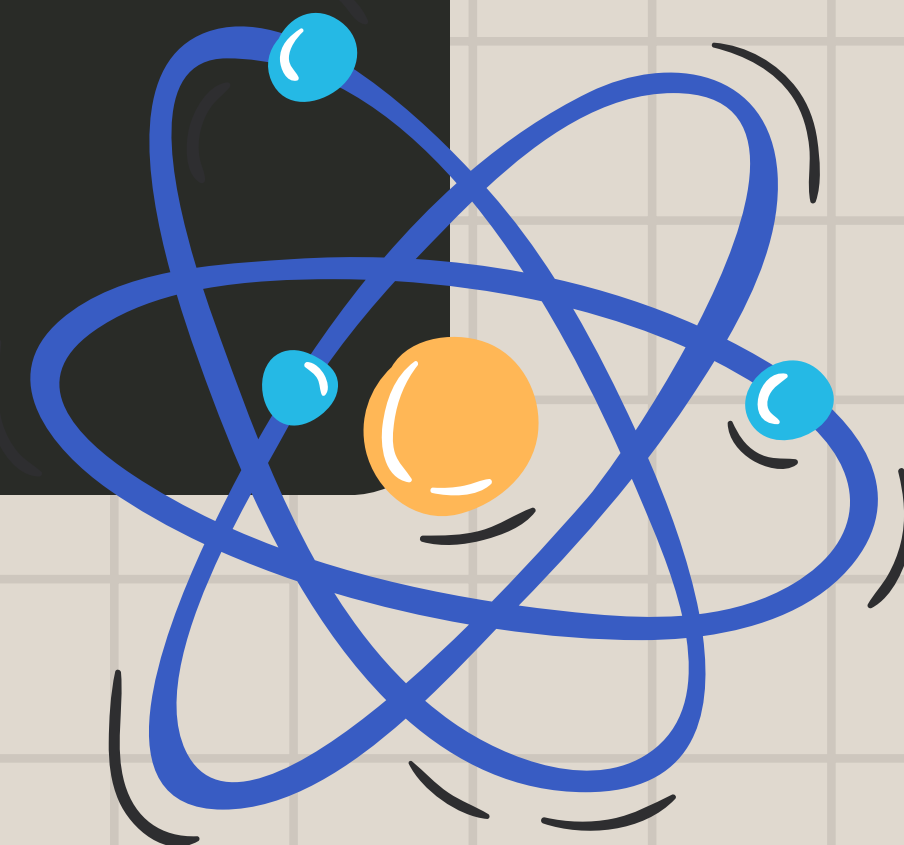




DENSITY DECODED

密度

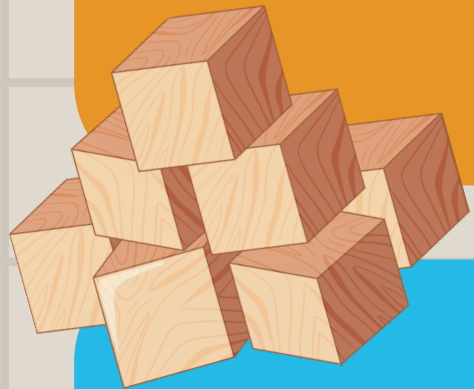


Rundown

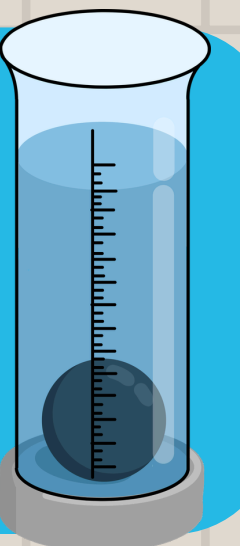
PART 1: WEIGHING AND CALCULATING



PART 2: FINDING THE MYSTERIOUS COMBINATION



PART 3: SHOWCASING THE ANSWER



PART 1: WEIGHING AND CALCULATING



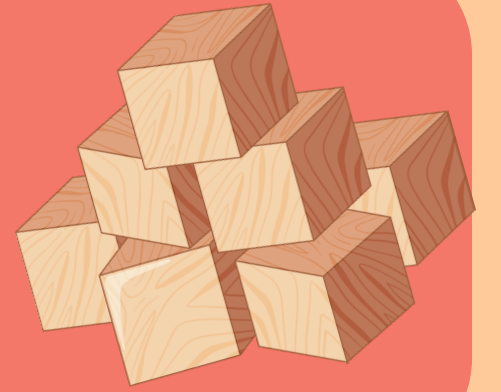
- There are 6 different blocks
- Density is provided
- Your goal:
 - Weigh the blocks
 - Calculate the volume (Calculator OK!)

PART 1: WEIGHING AND CALCULATING



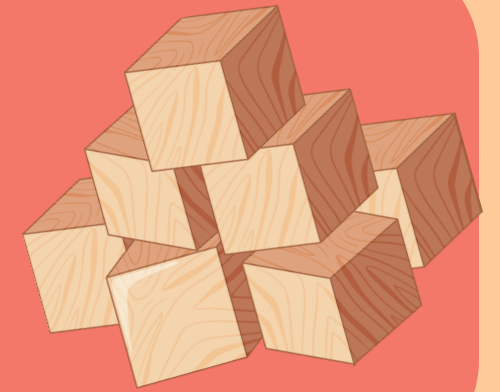
- ONLY one person may leave their seat from each group to go to the bench to weigh the blocks.
- The other students should use the data to calculate the volume of each block (Calculator can be used).
- 45 Minutes later, the worksheet will be collected.

PART 2: FINDING THE MYSTERIOUS COMBINATION



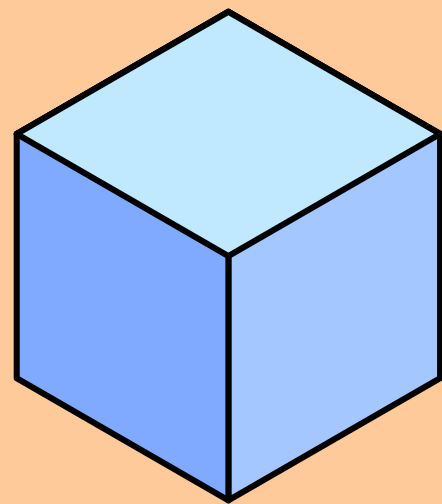
- A certain volume will be given on the screen
- Your goal: Figure out which combination can lead to that volume.
- **Special rule
 - One block **MUST BE** used **TWICE**
 - Other blocks are used **ONCE** or **ZERO** times

PART 2: FINDING THE MYSTERIOUS COMBINATION

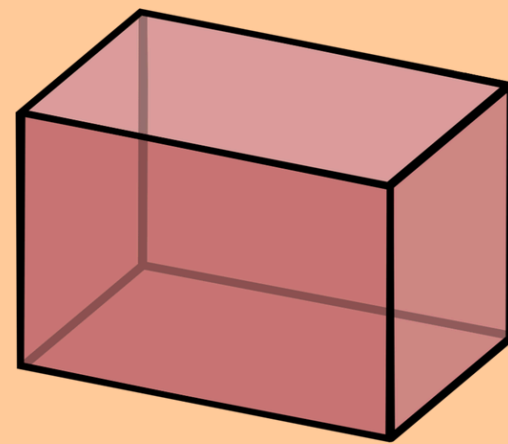


Example

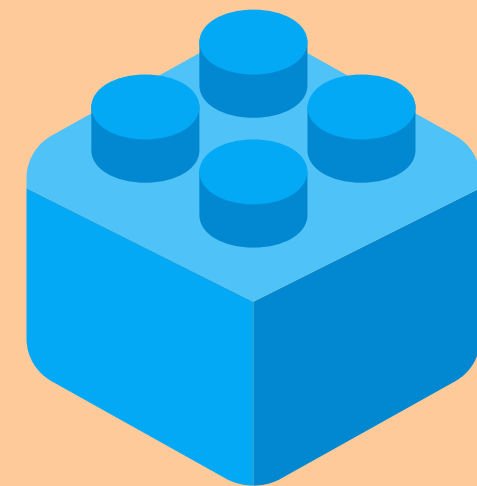
Target: 30 cm^3



12 cm^3

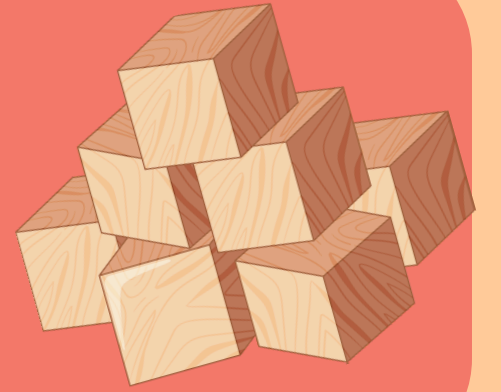


10 cm^3



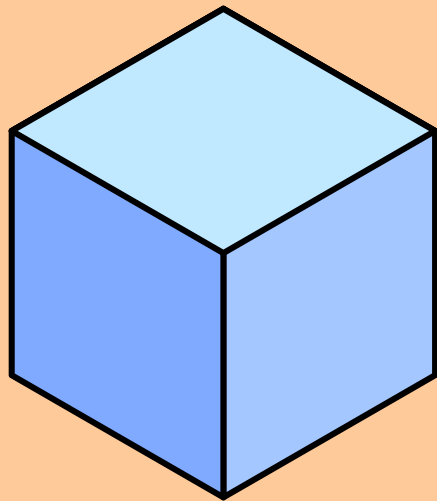
6 cm^3

PART 2: FINDING THE MYSTERIOUS COMBINATION



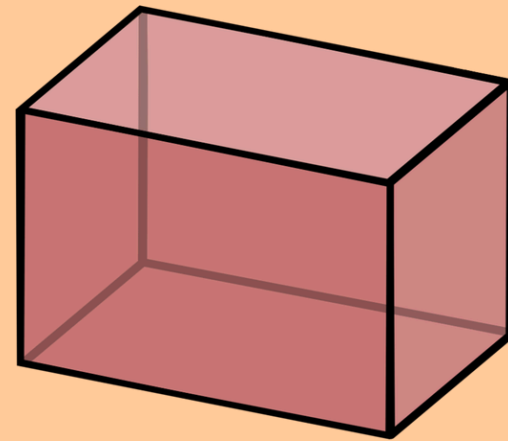
Example

Target: 30 cm^3



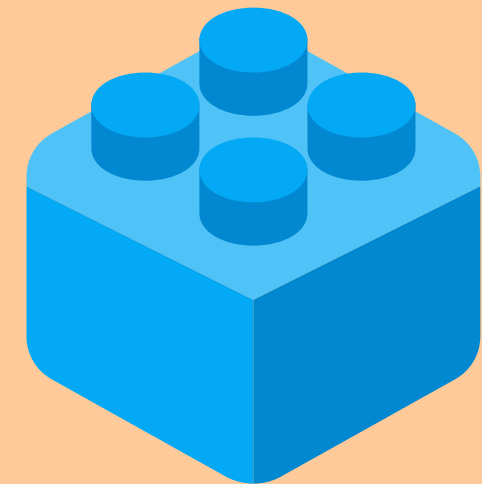
12 cm^3

x2



10 cm^3

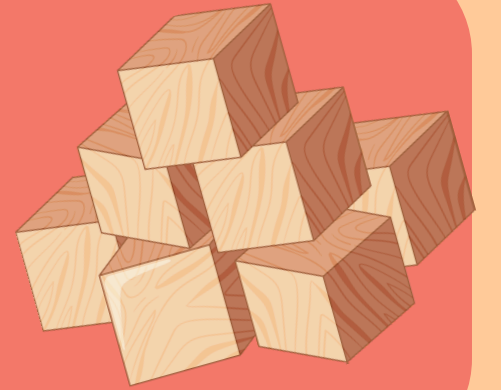
x0



6 cm^3

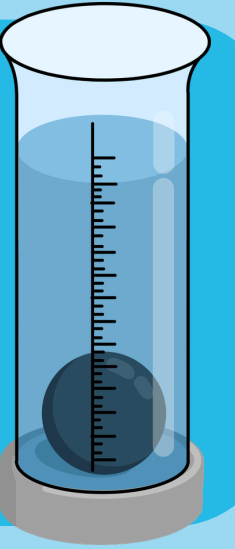
x1

PART 2: FINDING THE MYSTERIOUS COMBINATION



- **Special rule
 - One block *MUST BE* used *TWICE*
 - Other blocks are used *ONCE* or *ZERO* times
- There are *MULTIPLE ANSWERS*, no need to copy others!
- The worksheet will be collected after 45 minutes

PART 3: SHOWCASING THE ANSWER



- Water displacement will be used to demonstrate ALL the correct answers

MARK CALCULATIONS



PART B RESULT

- Groups who could get the combination correct is prioritized

PART A RESULT

- If there are multiple groups with correct combination, Part A results will be considered